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SUB-AMBIENT TEMPERATURE TRANSFER SYSTEM FOR COLD FORMING PROCESSES

Abstract

A system configured to roll a metal into a foil. The system includes work rollers spaced apart to accommodate the metal therebetween. The work rollers are configured to press against the metal to roll the metal into the foil. A cooling subsystem is spaced apart from the work rollers on a work line configured to feed the metal to the work rollers. The cooling subsystem is configured to cool the metal as the metal moves along the work line.

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Background/Summary

INTRODUCTION

[0001] The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

[0002] The present disclosure relates to sub-ambient temperature transfer systems for cold forming processes configured to roll metals into foils, the metals having relatively low melting temperatures.

[0003] Various manufacturing processes include rolling metals into foils. For example, lithium may be rolled into a thin foil to form an active layer of a battery anode. The lithium foil may be applied to any suitable current collector, such as a copper foil, to form the anode. Indium, tin, lead, and sodium are additional examples of metals that may be rolled into foils. In some cases, the metals may have a relatively low melting temperature.

SUMMARY

[0004] The present disclosure includes, in various features, a system configured to roll a metal into a foil. The system includes work rollers spaced apart to accommodate the metal therebetween. The work rollers are configured to press against the metal to roll the metal into the foil. A cooling subsystem is spaced apart from the work rollers on a work line configured to feed the metal to the work rollers. The cooling subsystem is configured to cool the metal as the metal moves along the work line.

[0005] In further features, the metal includes at least one of lithium, indium, tin, lead, and sodium.

[0006] In further features, the foil is configured as an active layer of a battery electrode.

[0007] In further features, the cooling subsystem includes cooling rollers defining channels therein configured to receive coolant to cool the cooling rollers and cool the foil in contact with the cooling rollers.

[0008] In further features, the channels extend entirely through the cooling rollers parallel to an axis of rotation of the cooling rollers.

[0009] In further features, the channels extend non-linearly relative to an axis of rotation of the rollers.

[0010] In further features, the channels include an inlet and an outlet at a common side of the cooling rollers.

[0011] In further features, the present disclosure includes polymeric support members between adjacent ones of the cooling rollers.

[0012] In further features, the cooling subsystem includes a housing adjacent to the cooling rollers, the housing defining a receptacle configured such that outer surfaces of the cooling rollers are accessible from within the housing and coolant present within the receptacle contacts the outer surfaces of the rollers. Gaps are defined between the housing and each one of the cooling rollers providing a rotational clearance for the cooling rollers and providing controlled release of the coolant from the receptacle onto the cooling rollers to cool and lubricate the cooling rollers.

[0013] In further features, adjacent rollers of the cooling rollers are offset from one another.

[0014] In further features, the cooling subsystem includes an upper guide and a lower guide opposite to the upper guide, the upper guide and the lower guide are spaced apart to accommodate the metal therebetween; the upper guide defines perforations configured to direct chilled air through the upper guide to the metal to cool the metal with an air blanket; and the lower guide is made of a thermally-conductive material and defines a cooling channel configured to circulate coolant through the lower guide to cool the metal, the air blanket pushes the metal against the lower guide.

[0015] In further features, the cooling subsystem includes an upper guide and a lower guide opposite to the upper guide, the upper guide and the lower guide are spaced apart to accommodate

the metal therebetween; the upper guide includes a guide roller made of an insulating material, the guide roller is configured to press the metal against the lower guide; and the lower guide is made of a thermally-conductive material and defines a cooling channel configured to circulate coolant through the lower guide to cool the metal, the guide roller is configured to push the metal against the lower guide.

[0016] In further features, the cooling subsystem includes an upper guide and a lower guide opposite to the upper guide, the upper guide and the lower guide are spaced apart to accommodate the metal therebetween; the upper guide defines perforations configured to direct chilled air through the upper guide to the metal to cool the metal with an air blanket; and the lower guide includes a heat pump configured to cool the metal, the air blanket pushes the metal against the lower guide.

[0017] In further features, the cooling subsystem includes an upper guide and a lower guide opposite to the upper guide, the upper guide and the lower guide are spaced apart to accommodate the metal therebetween; the upper guide includes a guide roller made of an insulating material, the guide roller is configured to press the metal against the lower guide; and the lower guide includes a Peltier stage configured to cool the metal, the guide roller is configured to push the metal against the lower guide.

[0018] In further features, the cooling subsystem includes a load guide defining a primary channel therethrough with the work line extending through the primary channel, a plurality of upper channels are above the primary channel, and a plurality of lower channels are below the primary channel; and the upper channels and the lower channels are configured to direct a coolant to the metal.

[0019] The present disclosure also includes, in various features, a system configured to roll a metal into a foil. The system includes: work rollers spaced apart to accommodate the metal therebetween, the work rollers configured to press against the metal to roll the metal into the foil; and a cooling subsystem spaced apart from the work rollers on a work line configured to feed the metal to the work rollers, the cooling subsystem including an upper guide and a lower guide on opposite sides of the work line, at least one of the upper guide and the lower guide is configured to be cooled with a coolant to cool the metal as the metal moves through the cooling subsystem.

[0020] In further features, at least one of the upper guide and the lower guide includes cooling rollers defining cooling channels configured for circulating the coolant through the cooling rollers.

[0021] In further features, one of the upper guide and the lower guide is configured to push the metal against the other of the upper guide and the lower guide.

[0022] The present disclosure further provides for a system configured to roll a metal into a foil. The system includes work rollers spaced apart to accommodate the metal therebetween, the work rollers configured to press against the metal to roll the metal into the foil; and a cooling subsystem spaced apart from the work rollers on a work line configured to feed the metal to the work rollers, the cooling subsystem including an upper guide and a lower guide on opposite sides of the work line, at least one of the upper guide and the lower guide is configured to cool the metal as the metal moves through the cooling subsystem. The metal includes at least one of lithium, indium, tin, lead, and sodium, and the foil is configured as an active layer of a battery electrode.

[0023] In further features, at least one of the upper guide and the lower guide includes a heat pump configured to cool the metal.

[0024] Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims, and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

[0026] FIG. 1 is a cross-sectional view of an exemplary battery cell including current collectors coated with active layers;

[0027] FIG. 2 illustrates an exemplary rolling system in accordance with the present disclosure for rolling a metal into a metallic foil;

[0028] FIG. 3A is a perspective view of an exemplary cooling subsystem in accordance with the present disclosure for cooling the metal;

[0029] FIG. 3B is a perspective view of exemplary cooling rollers of the cooling subsystem of FIG. 3A;

[0030] FIG. 3C illustrates additional exemplary cooling rollers in accordance with the present disclosure;

[0031] FIG. 3D illustrates alternative cooling rollers in accordance with the present disclosure;

[0032] FIG. 3E illustrates further cooling rollers in accordance with the present disclosure;

[0033] FIG. 3F illustrates additional cooling rollers in accordance with the present disclosure;

[0034] FIG. 4A is a perspective view of an additional cooling subsystem in accordance with the present disclosure;

[0035] FIG. 4B is a perspective view of an upper housing and a lower housing of the cooling subsystem of FIG. 4A, the housings in cooperation with cooling rollers;

[0036] FIG. 4C is a cross-sectional view of the upper housing and cooling rollers of FIG. 4B, the upper housing filled with a coolant;

[0037] FIG. 4D illustrates a clearance gap between the upper housing and one of the cooling rollers;

[0038] FIG. 5A is a perspective view of another cooling subsystem in accordance with the present disclosure;

[0039] FIG. 5B is a side view of an upper cooling guide and a lower cooling guide of the guide system of FIG. 5A;

[0040] FIG. 6 is a perspective view of another cooling subsystem in accordance with the present disclosure;

[0041] FIG. 7 is a perspective view of an additional cooling subsystem in accordance with the present disclosure;

[0042] FIG. 8 is a perspective view of a further cooling subsystem in accordance with the present disclosure;

[0043] FIG. 9 illustrates another cooling subsystem in accordance with the present disclosure;

[0044] FIG. 10 illustrates an additional cooling subsystem in accordance with the present disclosure; and

[0045] FIG. 11 is a side view of another exemplary rolling system in accordance with the present disclosure including a cooling subsystem with offset cooling rollers.

[0046] In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

[0047] The present disclosure includes a system configured to roll a metal into a foil. The metal may be any suitable metal having a relatively low melting temperature, such as one or more of lithium, indium, tin, lead, sodium, etc. The foil may be configured for use in any suitable automotive or non-automotive application. For example, the foil may be used as an active layer of a battery electrode. The battery may be configured for use with a vehicle, and may be configured for non-vehicular use as well.

[0048] Lithium, indium, tin, lead, sodium, and other metals with relatively low melting

temperatures lack mechanical strength at room temperature, which presents challenges with rolling such metals into foils. For example, such metals are prone to tearing and fracturing during rolling, and may stick to the work rollers. The present disclosure includes a cooling subsystem configured to reduce the temperature of the metals prior to reaching work rollers and/or between sets of work rollers. The system of the present disclosure provides for rolling of the metals at sub-ambient temperatures in a dry room environment with a dew point below the rolling temperature of the metals. Reducing the temperature of metals with relatively low melting temperatures provides the metals with increased strength and hardness, which allows the metals to be rolled to thinner foil gauges. The present disclosure chills the metal in various ways prior to reaching the work rollers, as described in detail herein.

[0049] FIG. 1 illustrates an exemplary battery cell **10**. The battery cell **10** may be configured for use in any suitable application, such as any suitable automotive or non-automotive application. The battery cell **10** includes C cathode electrodes **20**, A anode electrodes **40**, and S separators **32** arranged in a predetermined sequence in a stack **12**, which is seated in an enclosure **48**. C, A, and S are integers, which are each greater than one. In some examples, $A=C+1$. The C cathode electrodes **20-1**, **20-2**, . . . , and **20-C** include cathode active layers **24** arranged on one or both sides of cathode current collectors **26**. The A anode electrodes **40-1**, **40-2**, . . . , and **40-A** include anode active layers **42** arranged on one or both sides of the anode current collectors **46**.

[0050] With reference to FIG. 2, the anode active layers **42** may include a metal **112** rolled into a foil by a rolling system **110** of the present disclosure. The rolling system **110** includes a plurality of work rollers **120** arranged along a work line. The work rollers **120** are generally arranged in pairs spaced apart along the work line. The work rollers **120** of each pair are spaced apart to receive the metal **112** therebetween, and configured to press against the metal **112** to roll the metal **112** into a foil as the metal **112** is fed along the work line between the rollers **120** or each roller pair. The work rollers **120** may be made of any suitable material including, but are not limited to, steel, aluminum, copper, etc. The work rollers **120** may be formed in any suitable manner. For example, the work rollers **120** may be forged, cast, formed by additive manufacturing, etc. The additive manufacturing may be metal additive manufacturing followed by machining.

[0051] FIG. 3A illustrates an exemplary cooling subsystem **210** in accordance with the present disclosure configured to cool the metal **112**. The cooling subsystem **210** may be arranged at any suitable location along the work line of the rolling system **110** such that the cooling subsystem is in line with the work rollers **120**. For example, the cooling subsystem **210** may be arranged upstream of all of the work rollers **120** and/or between one or more pairs of the work rollers **120**. Multiple cooling subsystems **210** may be included so that a cooling subsystem **210** may be arranged between multiple different pairs of the work rollers **120**. For example, and as illustrated in FIG. 2, the cooling subsystem **210** may be arranged at one or more of the downward facing arrows of FIG. 2, as can any of the other cooling subsystems **310**, **410**, **510**, **610**, **710** described herein.

[0052] The cooling subsystem **210** includes a plurality of cooling rollers **220**, which are arranged both above and below the metal **112**. The cooling rollers **220** are spaced apart enough to not reduce the thickness of the metal **112** as the metal **112** passes across the cooling rollers **220** between upper and lower sets of the cooling rollers **220**. The cooling rollers **220** are configured to guide and transport the metal **112** along the work line to the work rollers **120**. With additional reference to FIG. 3B, between adjacent ones of the cooling rollers **220** are support members **222**. The support members **222** may be made of any material suitable to support movement of the metal **112** across the cooling rollers **220**, such as any suitable polymeric material. The support members **222** may be configured to generally provide a guide and table for the metal **112** to pass across.

[0053] The cooling rollers **220** are cooled by any suitable coolant circulated through cooling channels defined within the cooling rollers **220**. Cooling the cooling rollers **220** cools the metal **112** in contact with the cooling rollers **220**, which strengthens the metal **112** and allows the metal **112** to be rolled to relatively stronger and thinner foil gauges. Examples of coolant that may be used to

cool the cooling rollers **220** includes, but is not limited to, the following: silicone oil; liquid N.sub.2; liquid Ar; and liquid CO.sub.2.

[0054] The coolant may be cooled in any suitable manner, such as with any suitable chiller **650** (as illustrated in FIGS. **9** and **10**, for example). The chiller **650** may be any suitable chiller or other cooling device configured to cool a coolant to any suitable temperature, such as at least lower than an ambient or room temperature. The temperature may depend on the particular metal **112** being rolled by the rolling system **110**. For example, the temperature may be at or below the temperature at which the metal is subject to tearing or fracturing during rolling by the rollers **120**. More specifically, the chiller **212** may be configured to circulate coolant in a range of 25° C. to -70° C. Liquid Ar or liquid N.sub.2 are examples of coolants that may be used to achieve temperatures below -70° C. The metal being rolled may be at any suitable temperature. For example, the temperature of the metal being rolled may be from -80° C. to 25° C., such as -40° C. or about -40° C.

[0055] The cooling rollers **220** may be configured in various different ways to be in fluid cooperation with the chiller **650** and be cooled by the coolant. For example, and with reference to FIG. **3C**, the cooling subsystem **210** may include cooling rollers **220A** defining channels **230**. The channels **230** are configured to receive chilled coolant from the chiller **650**. The channels **230** are formed in any suitable manner, such as by being drilled into forged or cast solid cooling rollers **220A**. The channels **230** extend entirely through the cooling rollers **220A** along, or parallel to, an axis of rotation of the rollers **220A**. Each one of the channels **230** includes an inlet **232** and an outlet **234** on opposite ends of the cooling rollers **220A**.

[0056] FIG. **3D** illustrates additional exemplary cooling rollers **220B** in accordance with the present disclosure for use with the cooling subsystem **210**. The cooling rollers **220B** define channels **240**, which in the example illustrated extend entirely through the cooling rollers **220B** from inlets **242** to outlets **244** on opposite sides of the cooling rollers **220B**. The channels **240** are defined within the rollers **220B** during additive manufacturing of the rollers **220B**. The channels **240** extend non-linearly through the cooling rollers **220B** in generally a zig-zag fashion. The channels **240** are configured to receive chilled coolant from the chiller **650**.

[0057] With reference to FIG. **3E**, the cooling subsystem **210** may alternatively include cooling rollers **220C**. The cooling rollers **220C** are cast around tubes **250**, which may be made of copper or any other suitable material. The tubes **250** define channels for the coolant. The tubes **250** each include an inlet **252** and an outlet **254** on a same side of the rollers **220C**. The tubes **250** may alternatively extend completely through the rollers **220C**.

[0058] With reference to FIG. **3F**, the cooling subsystem **210** may alternatively include cooling rollers **220D**. The cooling rollers **220D** define channels **260**, which are generally porous areas of the rollers **220D** each having an inlet **262** and an outlet **264** on opposite sides of the channels **260**. The channels **260** are defined within the rollers **220D** during additive manufacturing of the rollers **220D**. The channels **260** are configured to receive the coolant.

[0059] FIGS. **4A-4D** illustrate an additional cooling subsystem **310** in accordance with the present disclosure for cooling the metal **112**. The cooling subsystem **310** is configured to direct coolant onto an exterior surface of the rollers **220**. The cooling subsystem **310** includes housings **320** on opposite sides of the metal **112**. The housings **320** include an upper housing and a lower housing. The housings **320** define receptacles **322**, which are configured to receive coolant, such as coolant chilled by the chiller **650**. The housings **320** are configured to support the cooling rollers **220** in any suitable manner that will allow the cooling rollers **220** to rotate and be exposed to coolant **324** within the receptacles **322** to cool the cooling rollers **220** and the metal **112** in contact with the cooling rollers, as illustrated in FIG. **4C**, for example. With reference to FIG. **4D**, gaps **326** are defined between the housing **320** and the cooling rollers **220** to allow the cooling rollers **220** to rotate and permit the coolant **324** to flow out from within the receptacle **322** and onto the cooling rollers **220**. The gap **326** may be any suitable size. For example, the gap **326** may provide a

rotational clearance of 0.1 mm-0.5 mm.

[0060] The coolant **324** may be any suitable liquid coolant, such as silicone oil or liquid argon, for example. The coolant **324** is in contact with the outer surface of the cooling rollers **220** to cool the cooling rollers **220**. The gaps **326** between the housing **320** and the cooling rollers **220** allow the coolant **324** to seep down onto the cooling rollers **220** to lubricate the cooling rollers **220**. When the coolant **324** is liquid argon, liquid argon fumes will leak from the gap **326** onto the cooling rollers **220** to provide an inert blanket to avoid moisture contamination on the cooling rollers **220**. Silicone oil and other inert cooling fluids also avoid side-reactions and any possible contamination of lithium.

[0061] FIGS. 5A and 5B illustrate an additional cooling subsystem **410** in accordance with the present disclosure. The cooling subsystem **410** includes an upper guide **420** and a lower guide **430**, which is opposite to the upper guide **420**. The upper guide **420** and the lower guide **430** are spaced apart to accommodate the metal **112** therebetween. The upper guide **420** defines perforations **422** configured to direct chilled air to the metal **112** from any suitable chilled air source. For example, the chiller **650** may be configured to chill the air. The upper guide **420** may be made of any suitable material, including any suitable metallic material, such as stainless steel. The upper guide **420** may be spaced apart from the metal **112** so as to not contact the metal **112**. The chilled air directed through the perforations **422** generates an air blanket **424**, which pushes the metal **112** against the lower guide **430**.

[0062] The lower guide **430** is generally a hollow enclosure made of any suitable thermally conductive material, such as copper or any other suitable thermally conductive alloy. The lower guide **430** serves as a load guide for the metal **112**, and may include any suitable coating to minimize friction between the metal **112** and the lower guide **430**, such as Teflon. Any suitable coolant is circulated through the lower guide **430** to cool the lower guide **430** and the metal **112** pressed against it. The coolant may be silicone oil, for example. As the metal **112** moves between the upper guide **420** and the lower guide **430**, the metal **112** is cooled by both the air blanket **424** and the lower guide **430**. The lower guide **430** may include an inlet **432** and an outlet **434** to circulate the coolant through the lower guide **430** from any suitable source, such as the chiller **650**. To facilitate movement of the metal **112** across the lower guide **430**, the lower guide **430** may include rounded edges on a downstream end of the lower guide **430** and an upstream end of the lower guide **430**. The lower guide **430** may also include any suitable coating, such as Teflon, to minimize friction between the metal **112** and the lower guide **430**.

[0063] With reference to FIG. 6, the upper guide **430** may be configured as a guide roll **440**. The guide roll **440** is configured to press the metal **112** against the lower guide **430** to facilitate cooling of the metal **112** by the lower guide **430**. The guide roll **440** may be made from any suitable insulating material. Although only one guide roll **440** is illustrated, the cooling subsystem **410** may include any suitable number of guide rolls **440**.

[0064] FIG. 7 illustrates an additional cooling subsystem **510** in accordance with the present disclosure for inclusion with the rolling system **110** to cool the metal **112**. The cooling subsystem **510** includes an upper cooling guide **520** and a lower cooling guide **530**. The upper cooling guide **520** is the same as, or substantially similar to, the upper cooling guide **420**. The upper cooling guide **520** defines perforations **522** configured to direct chilled air to the metal **112** to cool the metal **112** and generate an air blanket that presses the metal **112** into contact with the lower cooling guide **530**.

[0065] The lower cooling guide **530** is configured as a Peltier stage, which is configured to be cooled to sub ambient temperatures and configured as a load guide to support the metal **112**. The lower cooling guide **530** is generally a solid-state heat pump that operates on the Peltier effect. The lower cooling guide **530** includes an array **540** of p- and n-type semiconductor elements that are doped with electrical carriers. The semiconductor elements are arranged into the array **540**, which is electrically connected in series, and thermally connected in parallel. This array **540** is then

affixed to an upper ceramic substrate **550** and a lower ceramic substrate **552**, which are on opposite sides of the array **540** of the semiconductor elements. In general, heat is absorbed by the upper ceramic substrate **550** and transferred to the lower ceramic substrate **552**, which acts as a heat sink. Between the upper ceramic substrate **550** and the array **540** are copper conductors. Copper conductors are also included between the lower ceramic substrate **552** and the array **540**.

[0066] The p-type semiconductors of the array **540** are doped with certain atoms that have fewer electrons than necessary to complete the atomic bonds within the crystal lattice. When a voltage is applied, there is a tendency for conduction electrons to complete the atomic bonds. When conduction electrons do this, they leave “holes,” which essentially are atoms within the crystal lattice that now have local positive charges. Electrons are then continually dropping in and being bumped out of the holes and moving on to the next available hole. In effect, it is the holes that are acting as the electrical carriers. Electrons move relatively more freely in the copper conductors as compared to the semiconductors. When electrons leave the p-type and enter into the copper on the cold side at the upper ceramic substrate **550**, holes are created in the p-type as the electrons jump out to a higher energy level to match the energy level of the electrons already moving in the copper. The extra energy to create these holes comes by absorbing heat. Meanwhile, the newly created holes travel downwards to the copper on the hot side at the lower ceramic substrate **552**. Electrons from the hot side copper move into the p-type and drop into the holes, releasing the excess energy in the form of heat.

[0067] The n-type semiconductors of the array **540** are doped with atoms that provide more electrons than necessary to complete the atomic bonds within the crystal lattice. When a voltage is applied, the extra electrons are easily moved into the conduction band. However, additional energy is required to get the n-type electrons to match the energy level of the incoming electrons from the cold-side copper. The extra energy comes by absorbing heat. Finally, when the electrons leave the hot-side of the n-type semiconductors, they once again can move freely in the copper, drop down to a lower energy level, and release heat in the process. Thus, with the lower cooling guide **530** heat is absorbed at the cold side (upper ceramic substrate **550**) of the n- and p-type elements, and heat is released at the hot side (lower ceramic substrate **552**) of thermoelectric element to cool the metal **112**.

[0068] With reference to FIG. **8**, the upper cooling guide **520** may be configured as a guide roll **560**. The guide roll **560** is configured to press the metal **112** against the lower cooling guide **530** to facilitate cooling of the metal **112** by the lower cooling guide **530**. The guide roll **560** may be made from any suitable insulating material. Although only one guide roll **560** is illustrated, the cooling subsystem **510** may include any suitable number of guide rolls **560**.

[0069] FIG. **9** illustrates an additional cooling subsystem **610** in accordance with the present disclosure for inclusion with the rolling system **110** to cool the metal **112**. The cooling subsystem **610** includes a load guide **620** defining a primary channel **630** for the metal **112**, and a plurality of coolant channels **640** on opposite sides of the primary channel **630**. The coolant channels **640** are in fluid cooperation with the chiller **650** to receive chilled coolant from the chiller **650**. The coolant channels **640** are configured to direct coolant to the primary channel **630** to deposit the coolant on opposite sides of the metal **112**. The coolant may be any suitable coolant, such as silicone oil. The silicone oil chills the metal **112** and creates a lubricating film for moving the metal **112** through the load guide **620**. The coolant thus also acts as a lubricant to allow the metal **112** to move through the primary channel **630** without sticking to the load guide **620**, thereby enabling a continuous process. The load guide **620** may be made of any suitable material, such as any suitable polymeric material. The load guide **620** may be formed in any suitable manner, such as by any suitable 3D printing process. The metal **112** may be drawn through the load guide **620** in any suitable manner, such as within a vacuum. Thus, the cooling subsystem **610** may be sealed within any suitable air-tight chamber **660**. The vacuum also pulls off any residual moisture to keep the metal **112** inert.

[0070] FIG. **10** illustrates a cooling subsystem **710** in accordance with the present disclosure

configured to cool the metal **112** with a coolant in the form of chilled inert gas. The gas may be chilled in any suitable manner, such as by the chiller **650**. The cooling subsystem **710** includes a load guide **720** defining a plurality of machined channels **730** configured to direct chilled inert gas to the metal **112** to cool the metal **112**. The gas is applied under pressure to suspend the metal **112** within the load guide **720**.

[0071] FIG. **11** illustrates an exemplary cooling roller arrangement for any suitable cooling rollers **220**, such as the cooling rollers **220A**, **220B**, **220C**, etc. The metal **112** may be directed to the cooling rollers **220** by a guide roller **224**. The cooling rollers **220** are staggered or offset from each other, which results in a greater surface area of the metal **112** contacting the cooling rollers **220** and being cooled by the cooling rollers **220**. Thus, staggering the cooling rollers **220** generally increases the cooling efficiency of the cooling rollers **220**. The cooling rollers **220** may be staggered vertically as illustrated, or horizontally. The cooling rollers **220** are staggered with respect to the rotational axes thereof.

[0072] In the drawings, the systems of the present disclosure are generally illustrated as horizontal systems oriented to move left to right. However, the systems may be oriented in any suitable alternate orientation. For example, the orientation of the entire rolling mill setup (including the cooling rollers) may be changed from horizontal to vertical to take advantage of gravity and minimize the tensile (pulling) force induced on the thin Li foils during rolling. This may reduce the potential of a foil tear during manufacturing.

[0073] The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

[0074] Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including “connected,” “engaged,” “coupled,” “adjacent,” “next to,” “on top of,” “above,” “below,” and “disposed.” Unless explicitly described as being “direct,” when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

[0075] In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

Claims

1. A system configured to roll a metal into a foil, the system comprising: work rollers spaced apart to accommodate the metal therebetween, the work rollers configured to press against the metal to roll the metal into the foil; and a cooling subsystem spaced apart from the work rollers on a work line configured to feed the metal to the work rollers, the cooling subsystem configured to cool the metal as the metal moves along the work line.
2. The system of claim 1, wherein the metal includes at least one of lithium, indium, tin, lead, and sodium.
3. The system of claim 1, wherein the foil is configured as an active layer of a battery electrode.
4. The system of claim 1, wherein the cooling subsystem includes cooling rollers defining channels therein configured to receive coolant to cool the cooling rollers and cool the foil in contact with the cooling rollers.
5. The system of claim 4, wherein the channels extend entirely through the cooling rollers parallel to an axis of rotation of the cooling rollers.
6. The system of claim 4, wherein the channels extend non-linearly relative to an axis of rotation of the rollers.
7. The system of claim 4, wherein the channels include an inlet and an outlet at a common side of the cooling rollers.
8. The system of claim 4, further comprising polymeric support members between adjacent ones of the cooling rollers.
9. The system of claim 4, wherein the cooling subsystem includes a housing adjacent to the cooling rollers, the housing defining a receptacle configured such that outer surfaces of the cooling rollers are accessible from within the housing and coolant present within the receptacle contacts the outer surfaces of the rollers, and wherein gaps are defined between the housing and each one of the cooling rollers providing a rotational clearance for the cooling rollers and providing controlled release of the coolant from the receptacle onto the cooling rollers to cool and lubricate the cooling rollers.
10. The system of claim 4, wherein adjacent rollers of the cooling rollers are offset from one another.
11. The system of claim 1, wherein: the cooling subsystem includes an upper guide and a lower guide opposite to the upper guide, the upper guide and the lower guide are spaced apart to accommodate the metal therebetween; the upper guide defines perforations configured to direct chilled air through the upper guide to the metal to cool the metal with an air blanket; and the lower guide is made of a thermally-conductive material and defines a cooling channel configured to circulate coolant through the lower guide to cool the metal, the air blanket pushes the metal against the lower guide.
12. The system of claim 1, wherein: the cooling subsystem includes an upper guide and a lower guide opposite to the upper guide, the upper guide and the lower guide are spaced apart to accommodate the metal therebetween; the upper guide includes a guide roller made of an insulating material, the guide roller is configured to press the metal against the lower guide; and the lower guide is made of a thermally-conductive material and defines a cooling channel configured to circulate coolant through the lower guide to cool the metal, the guide roller is configured to push the metal against the lower guide.
13. The system of claim 1, wherein: the cooling subsystem includes an upper guide and a lower guide opposite to the upper guide, the upper guide and the lower guide are spaced apart to accommodate the metal therebetween; the upper guide defines perforations configured to direct chilled air through the upper guide to the metal to cool the metal with an air blanket; and the lower guide includes a heat pump configured to cool the metal, the air blanket pushes the metal against

the lower guide.

14. The system of claim 1, wherein: the cooling subsystem includes an upper guide and a lower guide opposite to the upper guide, the upper guide and the lower guide are spaced apart to accommodate the metal therebetween; the upper guide includes a guide roller made of an insulating material, the guide roller is configured to press the metal against the lower guide; and the lower guide includes a Peltier stage configured to cool the metal, the guide roller is configured to push the metal against the lower guide.

15. The system of claim 1, wherein: the cooling subsystem includes a load guide defining a primary channel therethrough with the work line extending through the primary channel, a plurality of upper channels are above the primary channel, and a plurality of lower channels are below the primary channel; and the upper channels and the lower channels are configured to direct a coolant to the metal.

16. A system configured to roll a metal into a foil, the system comprising: work rollers spaced apart to accommodate the metal therebetween, the work rollers configured to press against the metal to roll the metal into the foil; and a cooling subsystem spaced apart from the work rollers on a work line configured to feed the metal to the work rollers, the cooling subsystem including an upper guide and a lower guide on opposite sides of the work line, at least one of the upper guide and the lower guide is configured to be cooled with a coolant to cool the metal as the metal moves through the cooling subsystem.

17. The system of claim 16, wherein at least one of the upper guide and the lower guide includes cooling rollers defining cooling channels configured for circulating the coolant through the cooling rollers.

18. The system of claim 16, wherein one of the upper guide and the lower guide is configured to push the metal against the other of the upper guide and the lower guide.

19. A system configured to roll a metal into a foil, the system comprising: work rollers spaced apart to accommodate the metal therebetween, the work rollers configured to press against the metal to roll the metal into the foil; and a cooling subsystem spaced apart from the work rollers on a work line configured to feed the metal to the work rollers, the cooling subsystem including an upper guide and a lower guide on opposite sides of the work line, at least one of the upper guide and the lower guide is configured to cool the metal as the metal moves through the cooling subsystem; wherein the metal includes at least one of lithium, indium, tin, lead, and sodium, and the foil is configured as an active layer of a battery electrode.

20. The system of claim 19, wherein at least one of the upper guide and the lower guide includes a heat pump configured to cool the metal.
